Myungsin Cho

Teaching Statement

I have taught mathematics at different levels and in a range of formats. My experience spans general-education courses for non-majors, service courses such as calculus, proof-based linear algebra, and applied mathematics for STEM students. I have taught in traditional classrooms and large lectures with more than 100 students, as well as in flipped classrooms and synchronous online lectures.

Across these settings, my goal remains the same: to help students develop the ability to think mathematically for themselves. I approach mathematics as one connected story, organizing courses around questions that motivate new concepts and drawing attention to patterns of reasoning that recur across different subjects.

Teaching Mathematics as a Connected Whole

I design my courses around ideas that connect topics that might otherwise seem unrelated. In Calculus, I used approximation as a unifying theme. Throughout the semester, I returned to the ways limits, derivatives, and integrals develop this idea, giving students a common reference point as they encountered new material. When introducing a topic, I made clear what questions it answered and how it changed what they could do with the ideas they had already learned.

In proof-based Linear Algebra, I began with the familiar task of solving systems of linear equations, then shifted from finding particular solutions to understanding their underlying structure. Questions about existence, uniqueness, and classification motivated more abstract concepts such as linear transformations and subspaces. Students first had a reason to ask these questions, and only then developed the language needed to answer them systematically.

Across courses, I ask students to notice the questions and strategies of reasoning that repeatedly appear in mathematics. When teaching proofs, I emphasize their overall structure, encouraging students to identify what is given and what must be established before working through the intermediate steps. When reading a textbook proof, I ask students to consider how its author might have discovered the argument rather than simply following the finished text line by line.

Supporting Students' Learning

Different courses have different educational purposes, and I adapt my teaching accordingly. In Math of Beauty and Decision, a general-education course for nonmajors, I introduced mathematical ideas through familiar situations involving voting, symmetry, and games, helping students explore familiar situations from a mathematical perspective. In courses for STEM students, I place greater emphasis on building the mathematical foundations and reasoning skills they will need for further study, including understanding when and why to apply particular mathematical methods.

Students within the same course also have different levels of preparation and mathematical maturity. When someone has trouble following an argument, I ask where the reasoning stopped being clear and work from that point, often adding a more concrete example. For students seeking greater challenges, my lecture notes include material beyond what we cover in class, along with occasional questions in footnotes that invite further exploration. I also try to learn students' names and academic interests so that I can offer individual guidance during office hours.

In Analysis and Optimization, I have reordered some topics so that students encounter new concepts when they have a clearer reason to learn them. When I noticed that students with little experience writing proofs struggled to get started on assignments, I redesigned some homework problems as guided worksheets. These provide the overall structure of an argument while leaving key steps for students to develop and justify themselves. Students reported that the worksheets made proofs more accessible and intuitive.

I design assessments to reward consistent practice of fundamental skills while maintaining high expectations for conceptual understanding and mathematical reasoning. In Calculus, I allowed students to revisit missed exam problems for partial credit, giving them an opportunity to learn from their mistakes. As AI has made take-home work less reliable as a measure of independent understanding, I have placed greater emphasis on regular supervised assessments. In my current Analysis and Optimization course, homework provides low-stakes practice, and weekly quizzes draw directly on assigned homework problems. Quiz performance can count toward the computational part of the corresponding exam, allowing students to demonstrate mastery of fundamental skills through regular practice while exams devote greater attention to conceptual and proof questions.

Fostering Independent Mathematical Thinking

Explaining mathematics to another person is an important part of understanding it. In Linear Algebra, I used an oral final examination and provided sample problems and possible follow-up questions in advance. Preparing to explain their reasoning gave students an opportunity to identify what they understood and where they needed further work. During the examination, I could ask them to clarify a step or explain a choice, allowing me to assess their understanding beyond what a written answer alone might reveal.

The growing use of AI makes independent mathematical judgment increasingly important. I emphasize to students that using AI properly and responsibly requires more than effective prompting. They need enough mathematical understanding to examine assumptions, identify gaps in reasoning, and verify conclusions produced by these tools. I generally require students to submit homework by hand. Even when they have consulted external tools, writing out a solution gives them another opportunity to work through and verify the reasoning before submitting it.

My experience mentoring through a Directed Reading Program has also shaped how I think about undergraduate research. I encouraged students to raise their own questions and explain the material they studied, rather than simply complete readings and exercises. In undergraduate research, I would build on this approach by starting with accessible mathematical examples and helping students develop their observations into questions they can investigate. As a project progresses, I would give students greater responsibility for deciding what to explore next. Discovering new results would be rewarding, but I believe that learning to formulate and pursue mathematical questions is itself a valuable research experience.